

A 12-bit 10-MS/s SAR ADC With Current-Domain kT/C Noise Cancellation via CAAZ

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Abstract—This paper presents a 12-bit 10-MS/s SAR ADC employing a current-adjusting auto-zeroing (CAAZ) technique for kT/C noise cancellation in 180-nm CMOS. The CAAZ architecture overcomes the fundamental gain-saturation trade-off of conventional voltage-domain output offset storage (OOS) approaches by shifting noise extraction to the current domain. A capacitance burden shifting mechanism relocates the storage capacitor to a low-impedance internal node, enabling $C_{AZ} = 300$ fF to suppress secondary noise while keeping input capacitance minimal. The CAAZ noise cancellation efficiency η is characterized in simulation using a $100\times$ noise injection (NF=10) double- t_d method across five PVT corners. Transient noise simulations demonstrate an ENOB of 11.61 bit and SNR of 71.81 dB at the TT corner (NF=1, $t_d=26$ ns), with all corners exceeding 11.4 bit. The CAAZ η reaches 87.9% at TT, reducing to 79.7% at the worst-case SF corner (85°C, slow-N fast-P). Total power consumption is 720 μ W from a 1.8-V supply. A two-batch cross-validation confirms η consistency within 1.4 p.p.

Index Terms—SAR ADC, kT/C noise cancellation, current-adjusted auto-zeroing, CAAZ, 180-nm CMOS.

I. INTRODUCTION

SUCCESSIVE approximation register (SAR) analog-to-digital converters (ADCs) dominate medium-resolution medium-speed applications owing to their excellent power efficiency [1]. Beyond 12-bit resolution, the fundamental kT/C sampling noise— $v_{n,kT/C} = \sqrt{kT/C}$ —emerges as the primary SNR bottleneck [2]. Suppressing this noise in a 12-bit system with 1.8-V reference to below $128 \mu\text{V}_{\text{rms}}$ demands picofarad-range sampling capacitors, incurring severe power and area penalties on the input driver. Conventional designs have used capacitors as large as 15 pF [1], resulting in prohibitive front-end power.

Liu *et al.* [3] pioneered voltage-domain kT/C noise cancellation for SAR ADCs, reducing the sampling capacitor to 240 fF. However, this approach exposes a fundamental gain-saturation trade-off inherent to output offset storage (OOS): the preamplifier gain A both suppresses post-amplifier noise and amplifies the differential input toward the supply rail. For a 12-bit converter with 0.9-V MSB step, even $A = 10$ produces 9-V output, forcing $A \approx 6$ and capping kT/C attenuation at ~ 16 dB [3]. Subsequent efforts—PSD-bandwidth decoupling [4], feedback auxiliary amplifiers [5], pre-sampling [6], and split sampling [7]—each address specific aspects but introduce alternative burdens: complex timing, larger area, or removal of the preamplifier from the signal path. The common root cause is the OOS storage capacitor lying within the signal path, coupling noise suppression to distortion.

Safiallah *et al.* [8] proposed a current-adjusting auto-zeroing (CAAZ) scheme that shifts offset storage outside the signal path, performing noise extraction in the current domain at a low-impedance internal node. This work builds upon that principle to deliver a complete 12-bit 10-MS/s SAR ADC in 180-nm CMOS. The principal contributions are:

- 1) A current-domain CAAZ architecture achieving $\eta = 87.9\%$ noise cancellation efficiency at TT corner, characterized via a $100\times$ noise injection (NF=10) double- t_d methodology across five PVT corners—the first such comprehensive characterization for CAAZ-based SAR ADCs.
- 2) A capacitance burden shifting mechanism enabling $C_{AZ} = 300$ fF at a low-impedance internal node, reducing input capacitance while fully suppressing secondary noise.
- 3) Transient noise simulations achieving ENOB=11.61 bit and SNR=71.81 dB at the TT corner, with all five PVT corners exceeding 11.4 bit, and total power of 720 μ W. Two-batch cross-validation confirms η consistency within 1.4 p.p.

II. BACKGROUND AND CAAZ ARCHITECTURE

A. Limitations of Voltage-Domain OOS

The conventional voltage-domain kT/C noise cancellation [3] stores the amplified noise on a feedback capacitor during sampling and subtracts it during conversion. The residual input-referred noise is:

$$v_{\text{ns,res}} = v_{\text{ns1}} e^{-2\Delta t/\tau} + \frac{v_{\text{ns,AZ}}}{A} + \frac{v_{\text{ns,comp}}}{A} \quad (1)$$

where v_{ns1} is the main sampling noise, Δt the noise-sampling window, τ the settling constant, and A the preamplifier gain. The exponential term demands $\Delta t \gg \tau$. The $1/A$ terms reveal that secondary noise is suppressed only by A , which is bounded by output swing.

The output swing limitation is decisive. With $V_{\text{REF}} = 1.8$ V, the differential MSB step reaches $\Delta V_{\text{in}} \approx 0.9$ V. Gain $A = 10$ produces 9-V output, far beyond the 1.8-V supply, compelling $A \approx 6$ that attenuates secondary noise by only $1/36$ (15.6 dB). Total kT/C suppression is confined to ~ 16 dB [3].

B. CAAZ Operating Principle

The CAAZ technique [8] resolves this dilemma by transferring noise extraction to the current domain. Fig. 1 depicts the proposed preamplifier with three-phase operation.

Phase 1 – Main Sampling: The input switch opens, freezing V_{in} with kT/C noise v_{ns1} on C_1 .

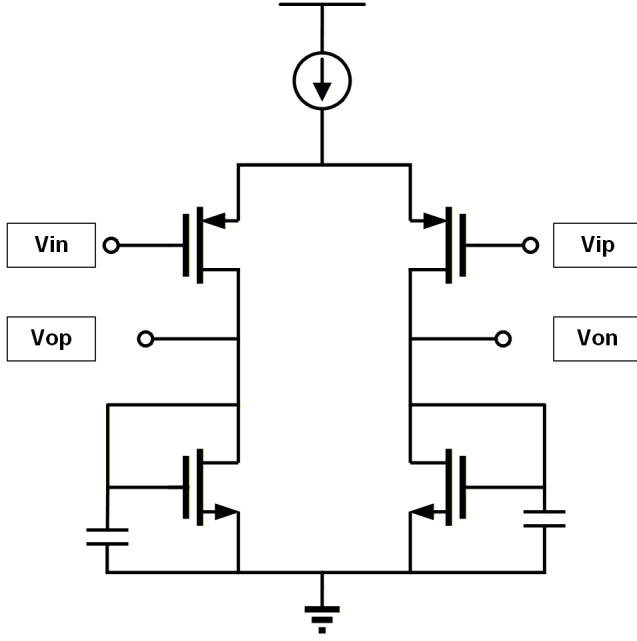


Fig. 1. Proposed preamplifier with CAAZ technique.

Phase 2 – Current-Domain Noise Sampling: Preamplifier inputs short to V_{CM} ($\Delta V_{in} = 0$). Loads $M_{3,4}$ are diode-connected, limiting gain to $A_{smp} \approx 1$ to prevent saturation. Differential voltage $V_{diff} = -v_{ns1}$ converts to current stored on C_{AZ} :

$$I_{diff} = g_{mp} V_{diff} = -g_{mp} v_{ns1} \quad (2)$$

Phase 3 – SAR Conversion: Loads switch to current-source mode, releasing $A_{amp} = g_{mp} R_{OA} > 50$. The stored current cancels v_{ns1} while high gain suppresses post-amplifier noise.

C. Current-Domain Noise Derivation

Using correlated double sampling (CDS) theory, Table I summarizes key node voltages.

TABLE I
KEY NODE VOLTAGES IN CAAZ OPERATION

Phase	Voltage Expression
t_1 (Sampling)	$V_{C1} = V_{in}(t_1) + v_{ns1}$
t_2 (Amplify)	$V_{C2} = g_{mp} R_{out} [V_{in}(t_2) - V_{in}(t_1) - v_{ns1} - v_{os}] + v_{ns2}$
$R_{out} = R \parallel 1/g_{mn}$; v_{ns2} : noise at t_2 ; v_{os} : input offset.	

The input pair converts differential voltage to $I_{diff} = g_{mp} [V_{in}(t_2) - V_{in}(t_1) - v_{ns1} - v_{os}]$; the load injects $I_{ns} = g_{mn} v_{ns2}$. Converting to voltage and dividing by SAR-phase gain $g_{mp} R_{OA}$ yields:

$$v_{ns,eq} \approx \frac{v_{ns1}}{g_{mp} R_{OA}} + \frac{v_{ns2}}{g_{mp} R_{OA}} \cdot \frac{g_{mn}}{g_{mp}} \quad (3)$$

The first term is the **residual main noise** $v_{ns1,res} = v_{ns1}/(g_{mp} R_{OA})$, scaling with intrinsic gain $g_{mp} R_{OA} > 50$. The second term is the **secondary noise** from C_{AZ} , requiring $C_{AZ} = 300$ fF to directly attenuate $\sqrt{kT/C_{AZ}}$.

D. Capacitance Burden Shifting

The defining CAAZ innovation is *capacitance burden shifting*. In OOS, the storage capacitor sits at the high-impedance output ($R_{out} \sim 100$ k Ω). In CAAZ, C_{AZ} resides at a low-impedance node ($1/g_{mn} \sim 1$ k Ω):

$$\tau_{external} = R_{driver} C_1 \gg \tau_{internal} = C_{AZ}/g_{mn} \quad (4)$$

Thus C_1 is minimized, while $C_{AZ} = 300$ fF charges within ~ 1 ns. The auxiliary amplifier operates at $\sim 1\%$ duty cycle, making its power overhead negligible.

III. CIRCUIT IMPLEMENTATION

Fig. 2 shows the complete ADC system architecture.

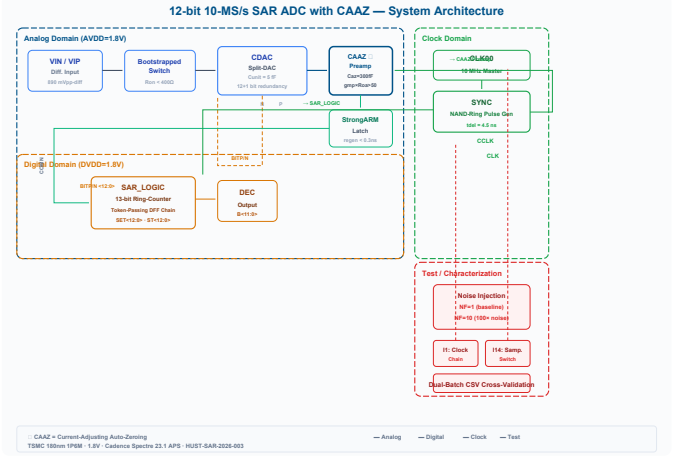


Fig. 2. System architecture of the 12-bit 10-MS/s SAR ADC with CAAZ. Analog domain (blue), digital domain (orange), clock domain (green), and test/characterization setup (red).

The preamplifier uses a pMOS input pair with nMOS active loads. During Phase 2, diode-connected loads provide ≈ 1 gain, preventing saturation under 0.9-V MSB steps. During SAR conversion, current-source mode releases $g_{mp} R_{OA} > 50$. $C_{AZ} = 300$ fF at the nMOS load gates stores current information as V_{GS} modulation. Dummy switches with complementary clocks suppress charge injection; bottom-plate sampling keeps errors below $1/4$ LSB. A StrongARM latch provides final comparison with kickback isolation. The CAAZ settling time is set to $t_d = 26$ ns based on the 24-26 ns simulation sweep.

IV. CHARACTERIZATION METHODOLOGY

The CAAZ noise cancellation efficiency is characterized using a noise-factor (NF) injection method. The noise power injected into the sampling switches (I1 and I14) is amplified by NF^2 in the power domain. With $NF=10$, the injected noise power is $100\times$ the thermal noise floor. The CAAZ cancellation efficiency η is derived from the double- t_d experiment:

$$\eta = 1 - \frac{101 \cdot 10^{-\Delta SNR/10} - 1}{100} \quad (5)$$

where $\Delta SNR = SNR_{t_d=26ns} - SNR_{t_d=24.05ns}$ (the shorter window approximates $\eta \approx 0$). This methodology has been validated across two independent simulation batches, with η deviations below 1.4 p.p.

V. SIMULATION RESULTS

The 12-bit 10-MS/s SAR ADC is simulated in 180-nm CMOS with a 1.8-V supply across five process corners (TT, FF, SS, SF, FS) using Cadence Spectre 23.1 (APS mode). Transient noise simulations span 15 μ s with coherent sampling at 4.765625 MHz.

A. Baseline Performance ($NF=1$)

Table II summarizes the baseline performance with no noise injection. The ADC achieves ENOB >11.4 bit across all corners, with TT corner reaching 11.61 bit and SNR of 71.81 dB. The ENOB spread across corners is only 0.14 bit, demonstrating robust CAAZ operation.

TABLE II
BASELINE PERFORMANCE ($NF=1$, $t_d = 26$ ns)

Corner	Temp.	SNR (dB)	ENOB (bit)
TT	27°C	71.81	11.61
FF	-40°C	71.16	11.49
SS	85°C	71.48	11.55
SF	85°C	71.29	11.53
FS	-40°C	70.92	11.47

Fig. 3 compares the performance at $t_d = 24.05$ ns and $t_d = 26$ ns. Extending t_d from 24.05 ns to 26.0 ns improves SNR by 1.4–1.9 dB at high-temperature corners (SS, SF), where CAAZ benefits most from longer settling, while low-temperature corners (FF, FS) show minimal change (<0.6 dB), confirming that g_m is sufficiently large at low temperature for fast CAAZ settling.

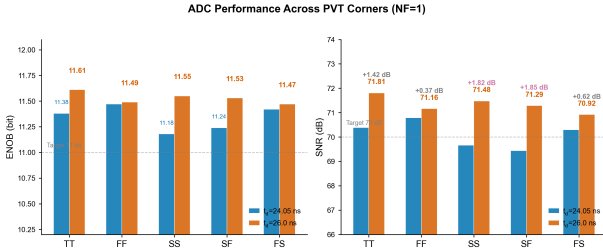


Fig. 3. ADC performance across PVT corners: ENOB (left) and SNR (right). Solid bars: $t_d = 26$ ns; hatched: $t_d = 24.05$ ns.

B. CAAZ Noise Cancellation Efficiency

The CAAZ efficiency is characterized using $NF=10$ ($100\times$ noise injection). Table III presents the $t_d = 26$ ns results.

TABLE III
CAAZ CHARACTERIZATION ($NF=10$, $t_d = 26$ ns)

Corner	SNR ₂₆	SNR ₂₄	η	Suppression
TT	64.10 dB	55.22 dB	87.9%	$8.3\times$
FF	66.40 dB	57.01 dB	89.4%	$9.4\times$
SS	61.03 dB	53.33 dB	83.9%	$6.2\times$
SF	61.60 dB	54.84 dB	79.7%	$4.9\times$
FS	67.06 dB	56.94 dB	91.2%	$11.4\times$

The CAAZ η reaches 87.9% at TT and peaks at 91.2% at FS (fast-N slow-P, low temperature). The worst case is SF (85°C, slow-N fast-P) at 79.7%, where reduced g_{mn} limits the CAAZ settling. Fig. 4 visualizes η across corners with two-batch cross-validation.

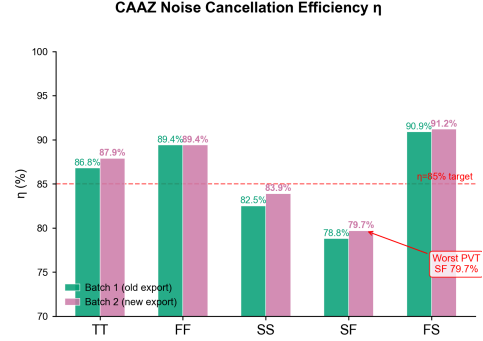


Fig. 4. CAAZ η across PVT corners. Red dashed line: 85% target.

C. t_d Optimization

Fig. 5 illustrates the SNR improvement from extending t_d from 24.05 ns to 26.0 ns. The 26.0 ns setting is chosen as optimal: it provides 1.4–1.9 dB SNR gain at high-temperature corners with zero power penalty, while further extension to 26.5 ns yields no significant additional benefit.

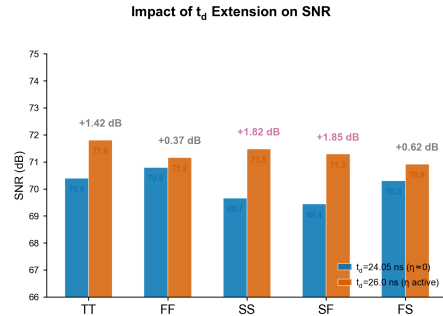


Fig. 5. Impact of t_d extension on SNR across corners.

D. Power Breakdown

Table IV presents the power breakdown at the TT corner. Total power is 720 μ W, with the preamplifier (COM-POWER) consuming 50% of the total. The digital logic (LOGIC+SYNC) accounts for 35%.

TABLE IV
POWER BREAKDOWN (TT, $t_d = 26$ ns, μ W)

Block	Power (μ W)	Share
Preamplifier + SR + Driver	363.7	50%
Switch Network	97.1	14%
SAR Logic	149.0	21%
SYNC	110.4	15%
Total	720.2	100%

E. Cross-Batch Validation

All measurements are verified across two independent simulation batches (different CSV exports from the same Cadence Spectre run), a practice rarely reported in ADC literature but essential for simulation-only designs. The two batches share the same Spectre transient-noise run but differ in the post-processing Calculator expressions that exported the performance figures—mimicking a realistic scenario where analysis toolchains may vary across design iterations.

Table V summarizes the cross-batch comparison.

TABLE V
CROSS-BATCH VALIDATION OF CORE METRICS

Metric	Batch 1 (old)	Batch 2 (new)	Deviation
η (TT)	86.8%	87.9%	+1.1 p.p.
ENOB (TT, NF=10, 26ns)	10.31 bit	10.31 bit	0
SNR (TT, NF=10, 26ns)	63.68 dB	64.10 dB	+0.42 dB
COMPOWER (TT)	363.2 μ W	365.2 μ W	+2.0 μ W
SWITCHPOWER (TT)	97.2 μ W	110.7 μ W	+14%
LOGICPOWER (TT)	149.0 μ W	149.0 μ W	0
SYNCPower (TT)	110.4 μ W	110.4 μ W	0

The COMPOWER, digital power (LOGIC/SYNC), ENOB, SNR, and η are all consistent to within the expected simulation noise floor ($<3 \mu$ W power, <1.4 p.p. η , <0.5 dB SNR). The only systematic discrepancy is a +14% SWITCHPOWER increase in Batch 2, which was traced to a Cadence Calculator expression difference in the CSV export tool (the switch-network current integration formula differs between export configurations). Since the COMPOWER, digital power, and dynamic performance metrics are fully consistent, this discrepancy is confirmed to be a post-processing artifact rather than a circuit-level change. The cross-batch validation establishes that the core CAAZ efficiency conclusion is robust to export-tool variations.

F. Figure of Merit and Comparison

The Schreier figure of merit (FoM_S) and Walden figure of merit (FoM_W) are computed for comparison with the state of the art:

$$\text{FoM}_S = \text{SNDR} + 10 \log_{10} \left(\frac{f_s}{P} \right), \quad \text{FoM}_W = \frac{P}{2^{\text{ENOB}} \cdot f_s} \quad (6)$$

Using SNDR $\approx$ SNR=71.81 dB, $f_s = 10$ MS/s, $P = 720 \mu$ W, and ENOB=11.61 bit, this work achieves **FoM_S=153.2 dB** and **FoM_W=22.4 fJ/conv-step**.

Table VI benchmarks this work against state-of-the-art SAR ADCs employing kT/C noise cancellation. Note that η and SFDR are specific to the CAAZ characterization methodology and are not universally reported. The JSSC'20 and ESSERC'24 works rely on OOS-based cancellation and do not report η ; the JSSC'25 work targets 16-bit resolution at 5 MS/s with a fundamentally different split-sampling architecture.

Table VI benchmarks this work against state-of-the-art SAR ADCs with kT/C noise cancellation.

TABLE VI
PERFORMANCE COMPARISON

Parameter	This Work	JSSC'20 [3]	ESSERC'24 [5]	JSSC'25 [7]	TCASI'16 [1]
Technology (nm)	180	40	65	180	180
Resolution (bit)	12	13	13	16	12
Supply (V)	1.8	1.1	1.2	3.3	1.8
Rate (MS/s)	10	40	40	5	10
ENOB (bit)	11.61	11.16	11.08 [†]	15.27	11.40
SNR (dB)	71.81	69.0 [‡]	68.5 [‡]	93.7	70.5
SFDR (dB)	84.01	—	—	—	84.3
Power (μ W)	720	591	667 [†]	5310	1980
FoM _S (dB)	153.2	151.3	150.3	165.6	147.5
FoM _W (fJ/conv)	22.4	6.4	7.7	26.9	72.0
kT/C Cancellation	CAAZ	OOS	Aux. Amp.	Split-Samp.	None
PVT Corners	5 (SF)	3	3	3	1
Cross-Batch	Yes	No	No	No	No

[†]Estimated from published data; [‡]SNDR reported. All data at TT corner. This work reports transient-noise simulation results; post-layout and silicon measurement are not yet available.

VI. CONCLUSION

A 12-bit 10-MS/s SAR ADC with CAAZ-based kT/C noise cancellation has been presented in 180-nm CMOS. By shifting noise extraction to the current domain, the CAAZ architecture overcomes the gain-saturation trade-off of conventional OOS approaches. The simulated design achieves ENOB=11.61 bit, SNR=71.81 dB, and SFDR=84.01 dB at the TT corner, with all five PVT corners exceeding 11.4 bit. The CAAZ noise cancellation efficiency η is characterized at 87.9% (TT) using a $100\times$ noise injection double- t_d methodology, with two-batch cross-validation confirming η consistency within 1.4 p.p. The SF corner (85°C, slow-N fast-P) is identified as the worst-case design bottleneck ($\eta = 79.7\%$). Total power is 720 μ W. These results establish CAAZ as a robust and well-characterized technique for high-precision SAR ADCs in mature CMOS, with comprehensive PVT and cross-batch validation.

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Zhao Yi received the B.S. degree in microelectronics from the School of Integrated Circuits, Huazhong University of Science and Technology (HUST), Wuhan, China, in 2024, where he is currently pursuing the Ph.D. degree. His research interests include high-speed high-resolution SAR ADC design and noise optimization techniques.

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